

Roy S Sigalingging

Full Stack Developer

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Profile

Information Technology graduate from Del Institute of Technology with a strong interest in Full Stack Development and professional experience as a Full Stack Developer at PT Citra Bonang, contributing to a web application migration from version 1.0 to 2.0 through feature development, system enhancement, and frontend-backend integration. Skilled in developing web applications using Laravel, React.js, and Next.js, building mobile applications with Flutter, managing MySQL databases, and integrating REST APIs, with additional experience in Machine Learning implementation. with strong problem-solving and teamwork skills, eager to learn new technologies and contribute to developing innovative and effective digital solutions.

Education

Del Institute of Technology September 2023 – September 2026
Associate Degree in Information Technology
GPA : 3.26 / 4.00

Work Experience

PT Citra Bonang February 2026 – July 2026
Full Stack Developer

Tech Stack: *CodeIgniter, PHP, jQuery EasyUI, JavaScript, MySQL*

- Collaborated in a team to migrate an inventory management application from version 1.0 to 2.0 using CodeIgniter.
- Developed backend functionality in CodeIgniter, including controllers, models, routing, and MySQL database queries, to support inventory data management.
- Built data-driven interfaces with jQuery EasyUI components such as DataGrid, forms, and dialogs to simplify the input, browsing, and management of inventory data.
- Optimized application performance and navigation to deliver a smoother user experience and faster load times.
- Worked with the internal team to refine business requirements and deliver production-ready features.

Project Experiences

Design and Development of the Rumah Damai Children's Foundation Information Website January 2024 – June 2024

Role: Full Stack Developer

Tech Stack: *Laravel, PHP, Blade Templates, HTML, CSS, JavaScript, MySQL*

- Developed a responsive information website for Rumah Damai Children's Foundation with Laravel.
- Built the front end with Blade Templates, HTML, CSS, and JavaScript to present foundation information clearly to visitors.
- Implemented backend functionality including application logic, routing, controllers, and database integration.
- Handled the full development cycle from design to deployment

Design & Development of the Pusmasib Mobile Application January 2025 – June 2025

Role: Full Stack Developer

Tech Stack: *Flutter, Dart, Laravel, PHP, MySQL, REST API*

- Developed a cross-platform mobile application with Flutter and Dart with smooth navigation and a clear user interface.
- Built the backend with Laravel, implementing business logic, routing, and controllers.
- Designed and managed the MySQL database schema for efficient data storage and retrieval.
- Connected the mobile app to the Laravel backend through REST APIs, enabling reliable data exchange between mobile and web systems.

Application of CNN Algorithm with MobileNetV4 Architecture for Cataract Classification August 2025 – January 2026

Role: *Machine Learning Engineer*

Tech Stack: *Python, PyTorch, MobileNetV4, CNN, Optuna, Flutter, Dart*

- Built a deep learning model with PyTorch and MobileNetV4 to classify eye images into three categories: Normal, Immature, and Mature cataract.
- Designed an end-to-end machine learning pipeline covering data preparation, training, validation, testing, and performance evaluation.
- Improved model performance through fine-tuning and hyperparameter optimization with Optuna (learning rate, batch size, dropout, weight decay, and classifier configuration).
- Deployed the model in a Flutter mobile application that lets users classify cataract conditions from eye images.

Mobile Application Development Project: Movie Catalog August 2024 – December 2024
Role:Mobile Application Developer

Tech Stack: *Flutter, Dart, REST API, SharedPreferences, SQLite*

- *Developed a movie catalog app with Flutter and Dart, integrating an external REST API to display movie data.*
- *Implemented state management, navigation flows, and local data persistence with SharedPreferences.*

Skills

- Programming Language: PHP, JavaScript, TypeScript, Dart, Python, Java, C, Golang
- Framework & Libraries: Laravel, CodeIgniter, React.js, Next.js, Flutter, Node.js, FastAPI, Flask, jQuery EasyUI, PyTorch, Optuna
- Testing Tools: Manual Testing, Selenium, Katalon
- Database Management: MySQL, PostgreSQL, MongoDB
- Clouds: Google Cloud Service
- Others: Git, GitHub, REST API, Postman, Figma, Android Studio, Trello

Course Taken & Certificate

- Toefl ITP - 467
- Google AI Essentials
- Share Data Through the Art of Visualization

Organizational Experience

- Facilities and Social Division of HIMATIF August 2024 – August 2026
- Del Data Science August 2024 – December 2024